

Assignment no. 1: Practical Assignments on Data Science Tools and Environment, Python Programming Basics

Objective: To develop familiarity with the data science tools and environment, understand the basics of Python programming for data science, work with Python data structures, and use data manipulation libraries like Pandas, NumPy, and Matplotlib.

Date: 10/07/2026

1.1 Introduction to the Laboratory Environment

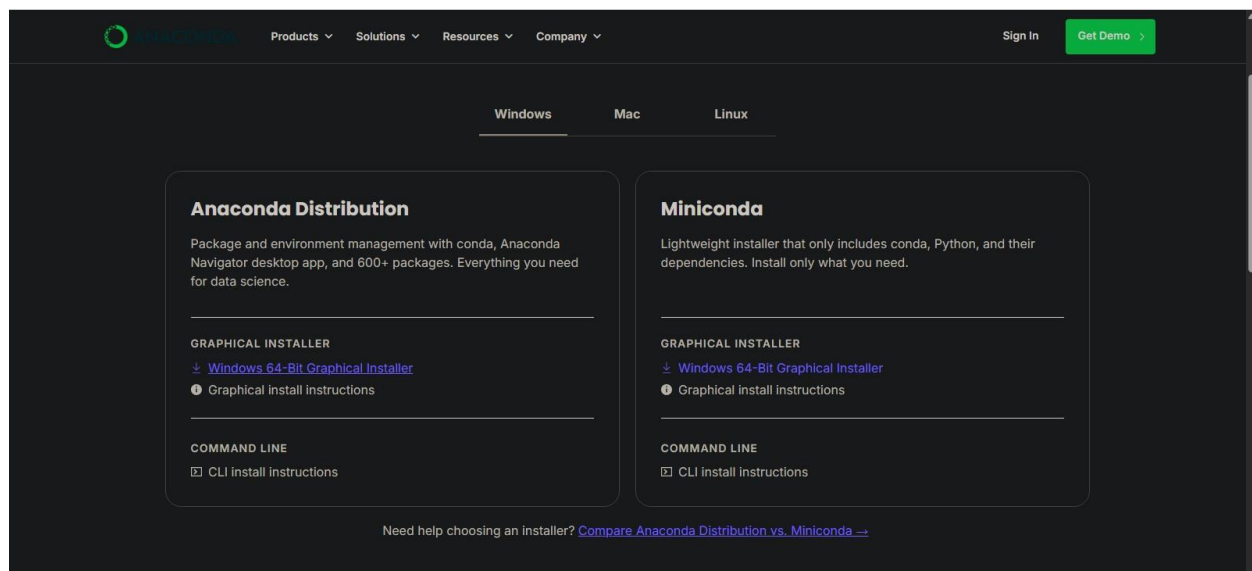
1. Installation and Setup:

o Install Anaconda distribution which includes Jupyter Notebook.

Step 1: Download Anaconda •

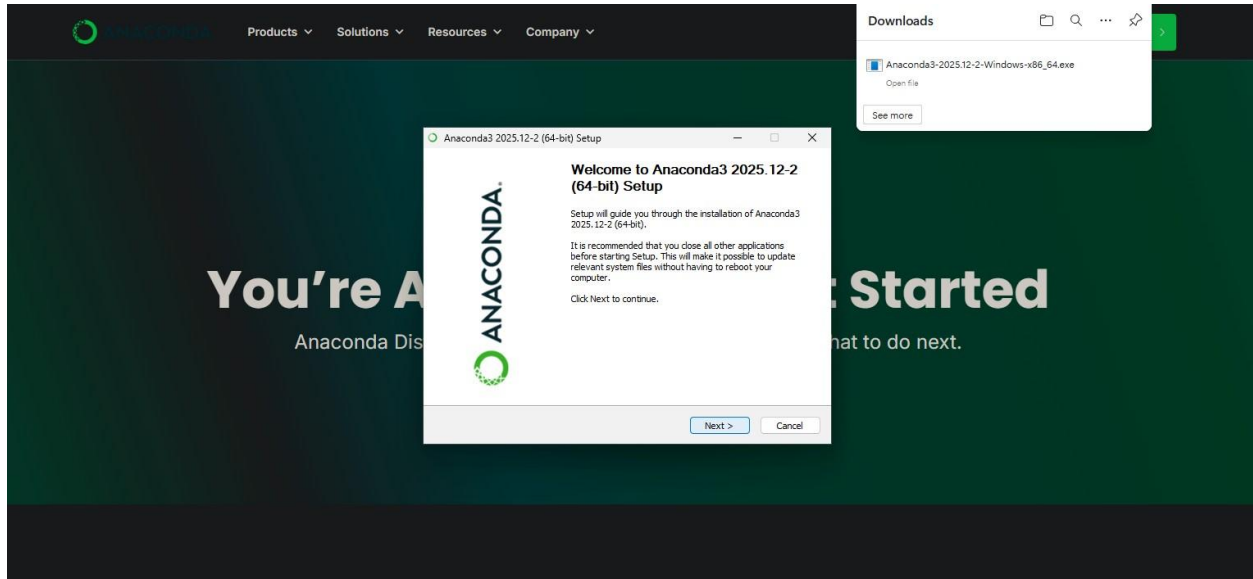
Open web browser.

- Visit: <https://www.anaconda.com/download> •
- Download - windows 64-bit-Graphical installer.

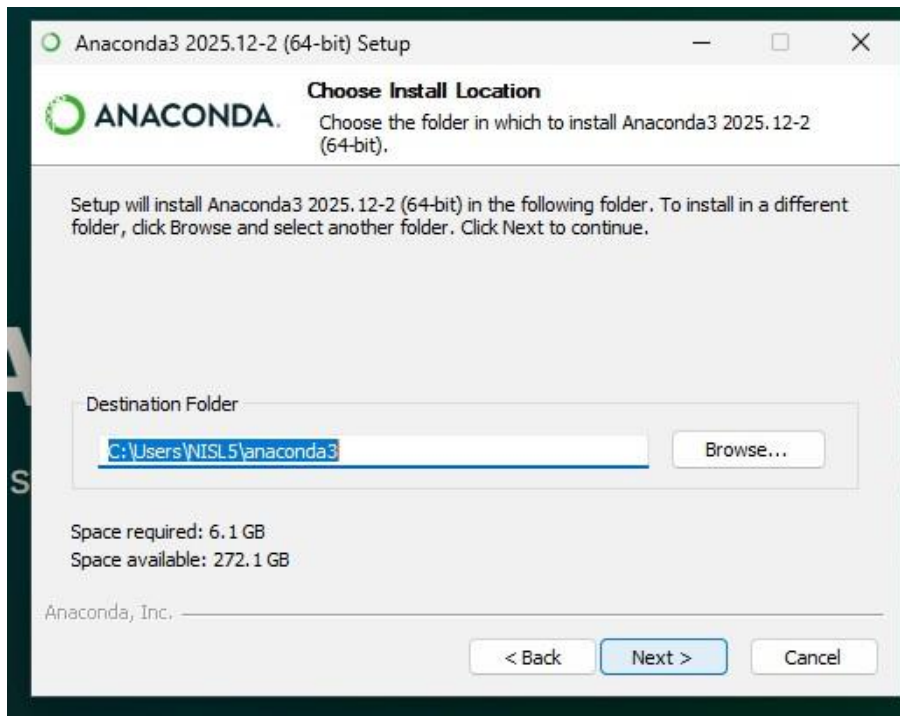


Step 2: Run Installer

- Double click the Installer
- Click on next



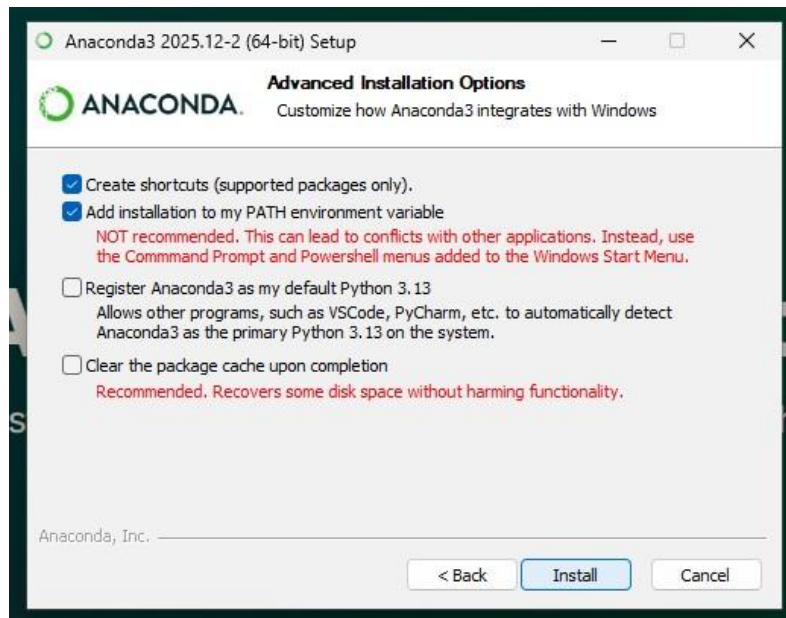
Step 3: Choose Installation location



Step 4: Advance Installation Options

- Select desired options •

Click on install



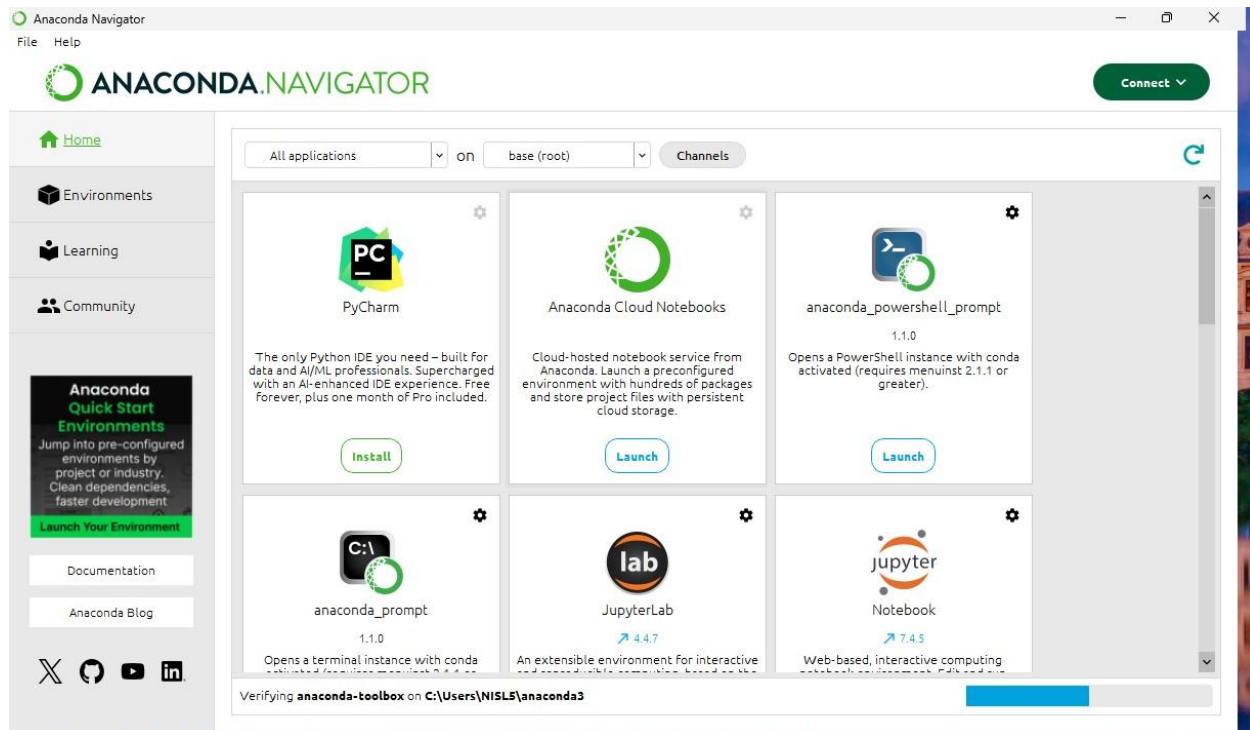
Step 5: Finish installation •

Click Finish



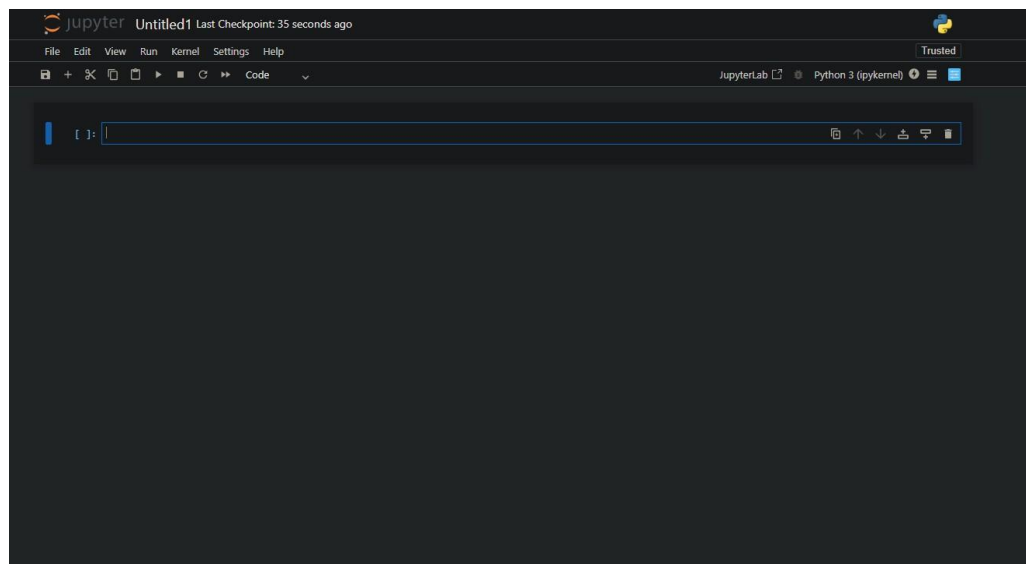
Step 6:

- Open Anaconda Navigator from start menu.



Step 7: Launching Jupyter notebook

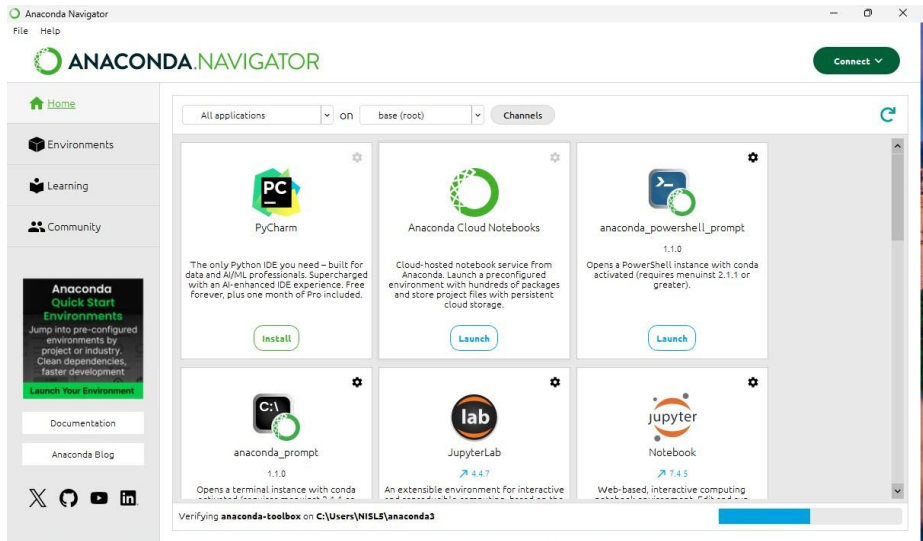
- Search for Jupyter Notebook
- Click on Launch
- Open the Notebook in new file and you are ready to go.



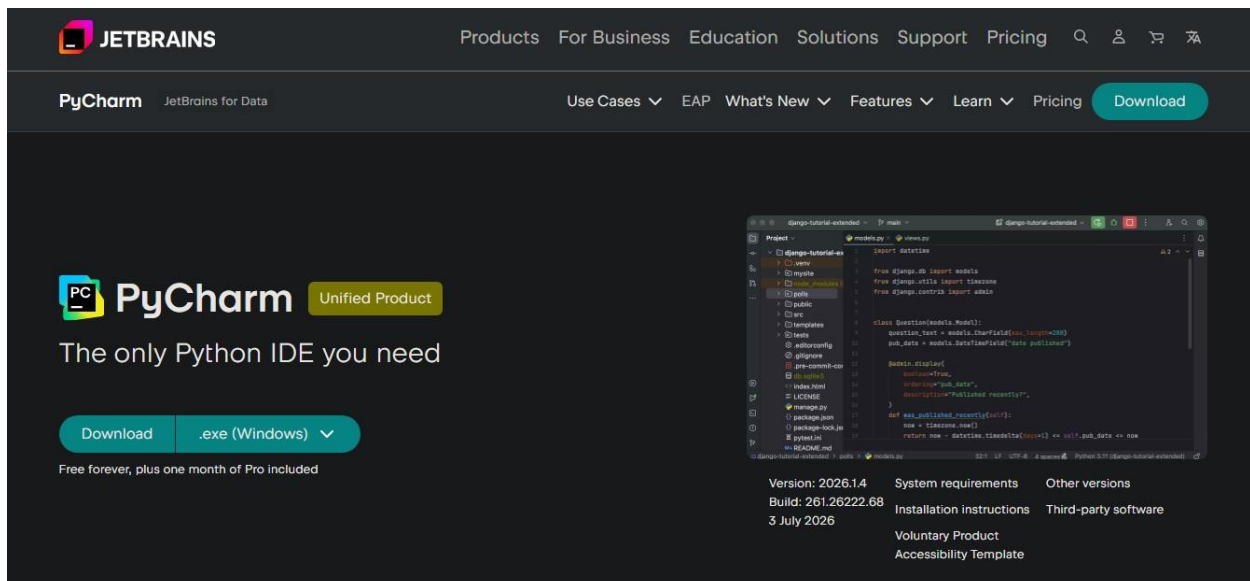
o Install a Python IDE

Step 1: Launching PyCharm

- Open Anaconda Navigator from start menu.
- Search for PyCharm
- Click on Install



Step 2: Download for windows

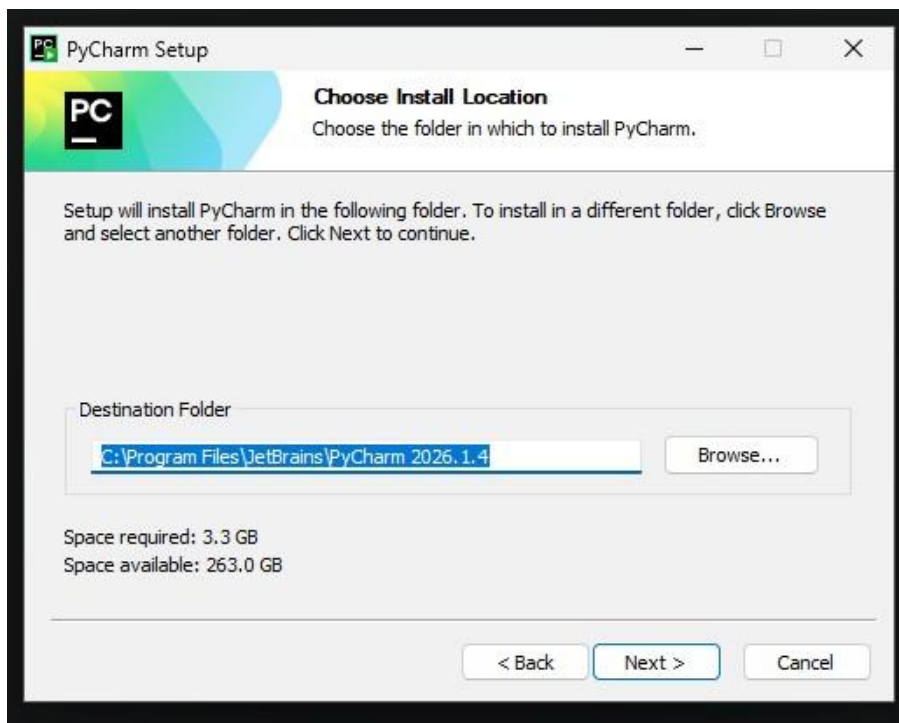


Step 3: Installer and Setup

- Open the installer • Click on next

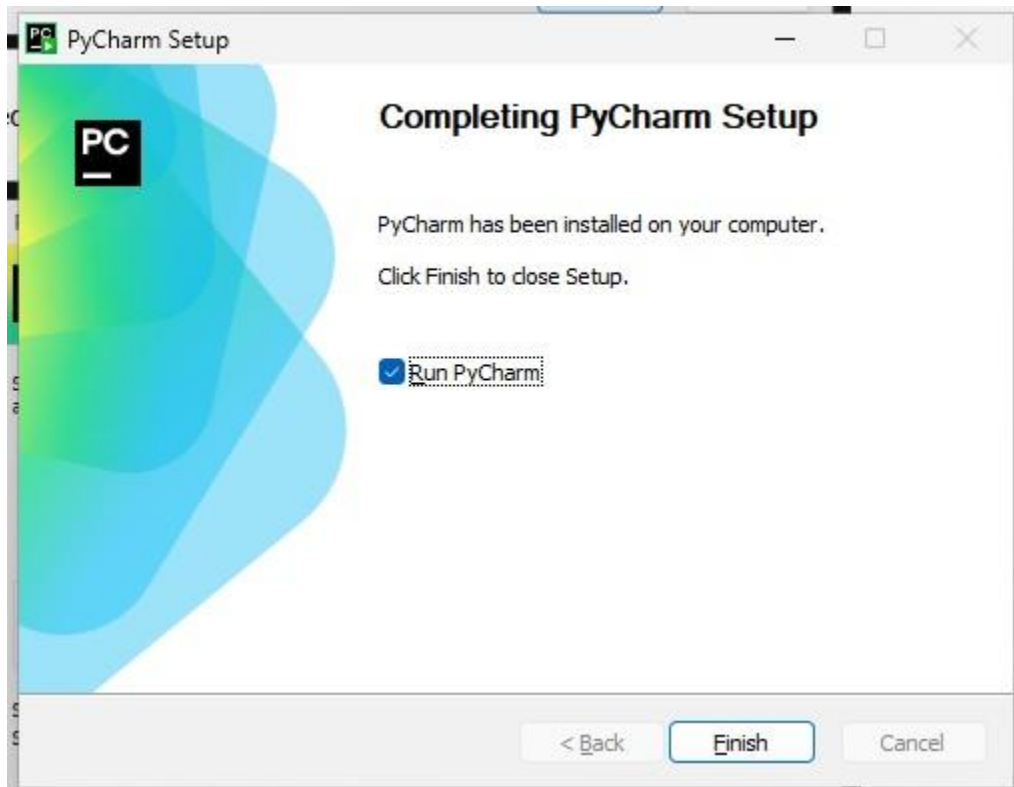


- Select installation location

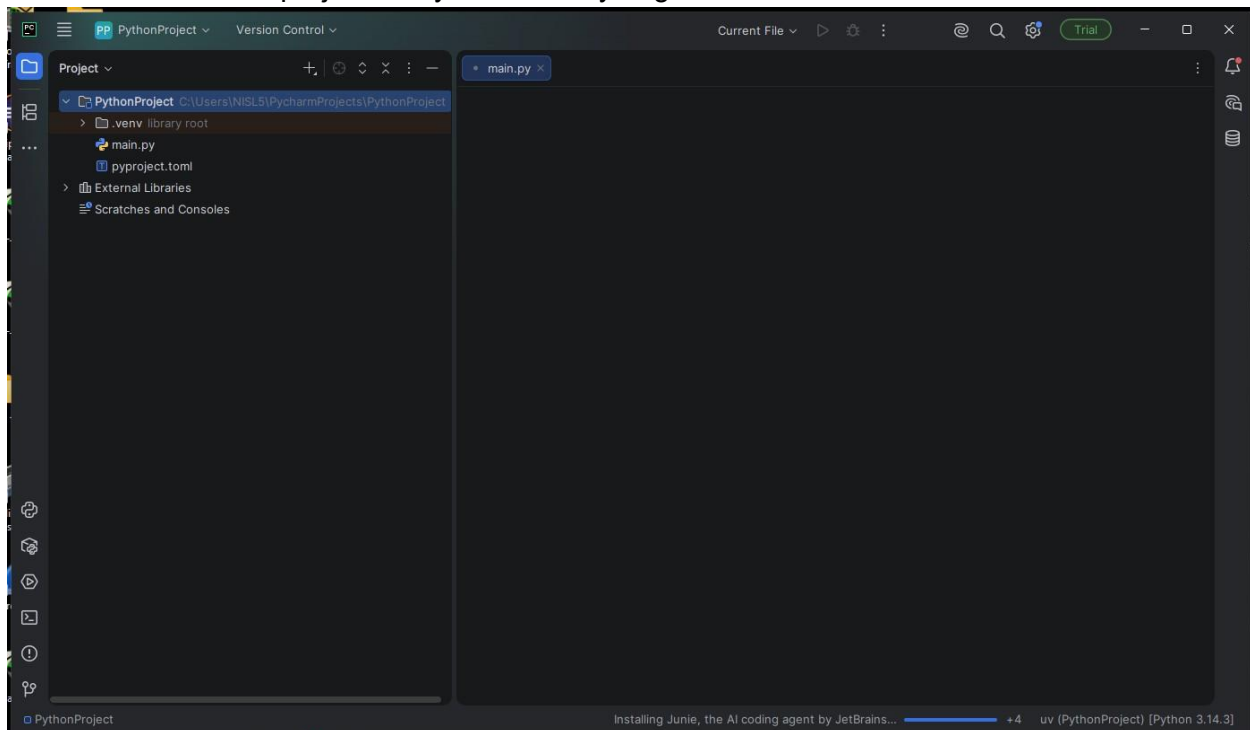


Step 4: Launch

- Click on Finish

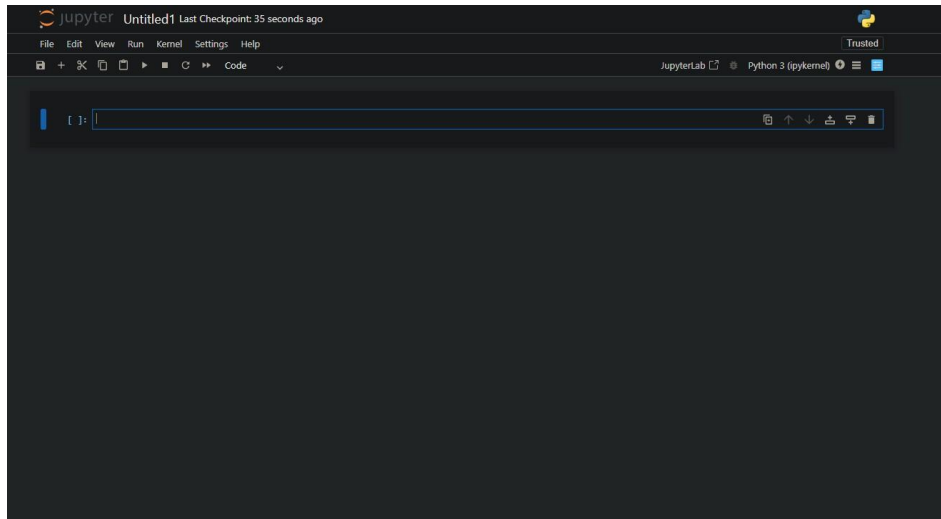


- Open PyCharm
- Click on New project and you are ready to go.



2. Jupyter Notebook Basics:

- o Create a new Jupyter Notebook.



- o Write a markdown cell explaining the purpose of the notebook and execute a simple Python code cell.

Data Science - Assignment 1

Purpose

This notebook demonstrates a simple Python program to check whether a number is even or odd.

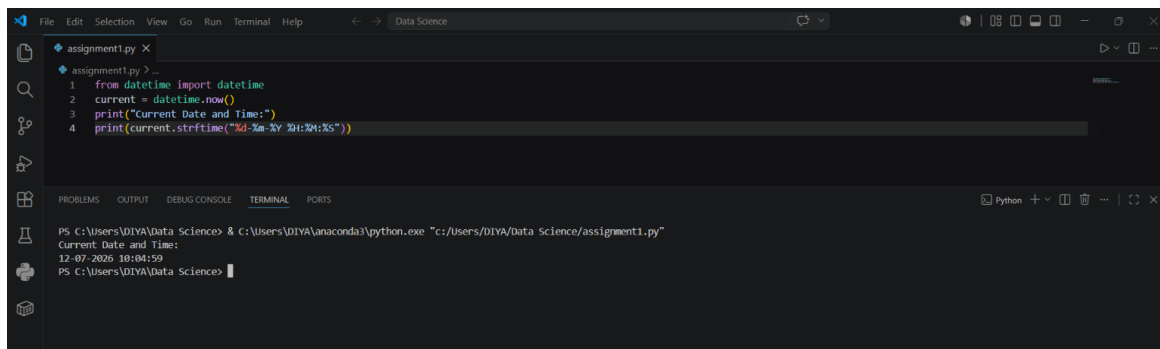
```
[1]: num = 8

if num % 2 == 0:
    print(num, "is Even")
else:
    print(num, "is Odd")

8 is Even
```

3. Exploring Python IDEs:

- o Write a simple Python script that prints the current date and time.



4. Comparing Tools:

- o Write a brief comparison (200 words) between Jupyter Notebook and the chosen Python IDE(VS code) based on your experience.

Ans:

As I am learning Python with the goal of exploring Data Science, I found that both the Jupyter Notebook and VS Code have their own strengths.

Jupyter Notebook provides an interactive environment where code is written and executed in separate cells. It also allows the use of Markdown cells to add explanations and notes alongside the code, making it easier to understand concepts step by step. This feature is especially helpful for beginners and for data analysis tasks where code, output, and documentation can be kept together.

In my first year of engineering, I used VS Code for writing programs in C, Java, and Python. It became familiar to me because of its features like syntax highlighting, code completion, integrated terminal, and debugging support. These features made it easier to identify errors and understand how programs execute, which was very useful while learning programming. After using both tools, I feel that each one has a different purpose. For my future journey in Data Science and Machine Learning, I think I will use the Jupyter Notebook more often because it is better suited for experimenting with data and documenting my work. At the same time, I will continue using VS Code for developing larger Python projects and strengthening my programming skills. Learning both tools has given me confidence and prepared me for different types of tasks.

1.2 Introduction to Data Science

1. What is Data Science?



Introduction:

Data science is the study of data used to extract meaningful insights for business decisions. It combines mathematics, computing and domain knowledge to solve real-world problems and uncover hidden patterns. It processes raw data to address business challenges and predict future trends. Big Data includes huge volume, high velocity, and extensible variety of data. There are 3 types: Structured data, Semi-structured data, and Unstructured data.

Steps involved in Data Science are:

Data Collection: Gathering raw data from various sources, such as databases, sensors or user interactions.

Data Cleaning: Ensuring the data is accurate, complete and ready for analysis.

Data Analysis: Applying statistical and computational methods to identify patterns, trends or relationships.

Data Visualization: Creating charts, graphs and dashboards to present findings clearly.

Decision-Making: Using insights to inform strategies, create solutions or predict outcomes.

Importance of Data Science:

Data science is important because it combines tools, methods, and technology to generate meaning from data. Modern organizations are inundated with data; there is a proliferation of devices that can automatically collect and store information. Online systems and payment portals capture more data in the fields of e-commerce, medicine, finance, and every other aspect of human life. We have text, audio, video, and image data available in vast quantities. Data science enables businesses to transform this data into strategic assets by:

Improving product recommendations and personalization

Detecting fraud and cybersecurity threats in real time

Enhancing decision-making at every level of the organization

Without data science, much of this data would remain untapped, offering little to no business value.

References:

<https://www.geeksforgeeks.org/data-science/data-science/>

<https://aws.amazon.com/what-is/data-science/>

<https://www.denodo.com/en/glossary/data-science-definition-importance-key-components>

2. Application of Data Science:

→

1. **Data Science in Marketing & Advertising:**

Data science applications have transformed marketing by creating personalised customer interactions and improving campaign results.

Customer journey mapping-

Customer journey maps show visual representations of consumer experiences with brands at every touchpoint. These maps track interactions, emotions and potential issues throughout the relationship. Good mapping leads to real business results, companies that deliver superior customer experience generate up to three times greater returns to shareholders.

The Customer Journey Mapping Process involves collecting data on your customer interactions, analysing their behaviour, identifying touchpoints, creating customer personas, and visualizing the journey map to gain actionable insights.

Once you have gathered data on customer interactions, it is essential to analyse their behaviour patterns. This analysis is crucial for understanding the various touchpoints where your customers engage with your brand. By identifying these touchpoints, you can pinpoint critical moments in the customer journey that significantly impact their experience.

Subsequently, creating detailed customer personas based on the data you have collected will assist you in personalizing the journey map. Visualizing the journey map with the integrated customer personas provides a comprehensive view that helps in recognizing pain points, identifying opportunities for improvement, and implementing strategic changes.

2. **Data Science in Education:**

Schools and colleges now use data science to boost learning outcomes with evidence-based methods. This new approach has changed how teachers understand their students' needs and teach their classes.

Student performance tracking-

Teachers can now watch how each student progresses over time and spot patterns in their academic work. Data science helps teachers predict which students might struggle and step in before grades drop. Schools look at test scores, attendance, and class participation to find learning gaps early and give extra help where needed.

Core Data Science Workflows

Exploratory Data Analysis (EDA): Educators use tools like Kaggle datasets and Python libraries (Pandas, Plotly) to visualize correlations—for instance, noting how a drop in attendance percentage directly impacts a student's total score.

Predictive Modelling: Machine Learning algorithms (Random Forest Regressors, Logistic Regression) are trained on historical data to predict if a student is at risk, allowing for early academic interventions.

Adaptive Learning & Recommendations: Advanced systems track skill proficiencies and cross-reference them with current job markets, recommending personalized courses or learning paths.

3.Data Science in Finance & Banking:

Data science adoption in banking has reshaped how financial institutions protect assets and run operations. Banks use sophisticated algorithms that have changed how they function in the digital world.

Credit scoring and loan approvals-

Data science has changed how lending decisions are made through better credit risk assessment. AI-powered credit scoring systems look at alternative data sources, rental payments, mobile data, and social media activities, to get a full picture of applicant creditworthiness. These systems process loans faster while maintaining deep analysis.

Different algorithms serve distinct purposes in credit risk assessment:

Logistic Regression: Often used as a baseline, this algorithm calculates the mathematical probability of a default (0 to 1) and is favoured for its transparency and interpretability.

Random Forest & Gradient Boosting: These ensemble tree-based models excel at capturing complex, non-linear relationships in data (e.g., how an applicant's age interacts with their specific job type and transaction history).

Deep Learning & Neural Networks: Applied to massive datasets to spot hidden risk signals, though these require extra steps to remain compliant with lending transparency laws.

References:

<https://talentsprint.com/blog/applications-of-data-science>

<https://www.dataquard.com/blog/what-is-customer-journey-mapping/>

3. Tools/Libraries for Data Science:



Tools for data science:

We're running in a world full of the digital era, and we've got a lot of data all over, and we are using different tools and techniques to make it resourceful for different purposes. To perform certain activities requires an individual to have proficiency in any of the programming languages (from a certain list) and various tools too.

1. SAS

Within Data Science, there are certain categories from which "data visualization" is also one of them. When you're working with visualization, SAS is the name of a tool that should come first in the "statistical" category. It is being used to create and present a symmetric chart of analytics and helps in managing data. For that, it uses the SAS programming language to outcast statistical modelling. If you're into this field or trying to get into data science, we recommend you learn this tool because most of the companies are working on relatable metrics and would require you to be familiar with these kinds of tools. Having a basic understanding of data science is what you need to become proficient in these tools and technologies.

It provides a list of features that will help you in deciding its application within your work, some of them are:

- Location Analytics
- Text Analytics
- Business Intelligence
- Augmented Analytics

2. Snowflake

Data Warehousing is one of the most important subsidiaries in Data Science today and the best tool to perform this action is the snowflake that's built on SQL for the cloud. The best part about this is it offers free hands in flexibility and efficiency which is not that easy without any big data platform. Besides this, it also offers some other benefits that make it more reliable for using it as a data warehousing tool, some of them are:

- The data cannot be accessed by anyone that's stored within it, in fact, it will require the user to access it via SQL query using snowflake to view the stored data.
- It builds all of its resources to create a virtual warehouse that is able to secure and process it even more fluently. This makes it to replicate the data across the cloud when and if there's any failure (to ensure a smooth business run).

- A user can access only a limited data as they'll pay which means every time, they want to access the data (or any portion) they'll have to pay a certain amount to do that.

Python libraries for data science:

Python libraries give data scientists access to a range of tools to help them manipulate, analyse, mine, and visualise data in a simple, straightforward manner. Each Python library contains sets of code, classes, values, and templates you can download to add functionality to Python to make analysing data more efficient.

1.NumPy:

NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object and tools for working with these arrays. It is the fundamental package for scientific computing with Python. Besides its obvious scientific uses, NumPy can also be used as an efficient multi-dimensional container of generic data.

Install commands: `conda install numpy` or `pip install numpy`

2.Pandas:

Pandas is a popular open-source Python library used for data manipulation and analysis. It provides powerful tools for working with structured data and integrates seamlessly with libraries such as NumPy and Matplotlib. Pandas operates around two core data structures:

Series (1D): Used to handle single columns of data efficiently.

DataFrame (2D): Used for structured, tabular data similar to spreadsheets or SQL tables.

Install commands: `conda install -c conda-forge pandas` or `pip install pandas`

pandas use expressive data structures to make working with labelled data more efficient. It simplifies data analysis in Python by representing missing data, allowing insertion or deletion of data, and converting data into data frames, which include merging, joining, and concatenate options. It also has useful In/Out (IO) functionality which you can use to import data directly from CSV, Excel, and databases.

3.Matplotlib:

Matplotlib is an essential tool for data science visualisations because it creates various data plots and graphs in print-ready formats. It creates plots like pairwise data, statistical graphs, gridded data, irregular data, and 3D volumes. Matplotlib works with Python scripts, Jupyter Notebook, web applications, and other graphic user interfaces (GUI) to generate plots, which makes it a versatile visualisation tool for data scientists.

Install commands: `conda install matplotlib` or `python -m pip install -U matplotlib`

References:

<https://www.coursera.org/in/articles/python-libraries-for-data-science>

<https://www.geeksforgeeks.org/data-science/data-science-with-python-tutorial/>

<https://www.geeksforgeeks.org/blogs/best-tools-and-technologies-for-data-science/>

4. Case Study:**A Case Study on Prediction of Google's Stock Price:****Abstract :**

Stock price prediction is a challenging task due to the dynamic and uncertain nature of financial markets. Traditional forecasting methods often provide limited accuracy, leading researchers to explore machine learning and deep learning techniques. This case study examines the use of a Long Short-Term Memory (LSTM) model to predict stock price trends using historical Google stock data. The study highlights the model's working process, evaluates its prediction accuracy, and demonstrates the potential of data science in supporting informed investment decisions.

Introduction:

Stock price prediction is a challenging yet important task in financial markets, as accurate forecasts help investors make informed decisions and reduce investment risks. Traditional methods, such as technical and fundamental analysis, often have limited accuracy due to market uncertainty. With advancements in data science, machine learning techniques like Long Short-Term Memory (LSTM) networks have improved prediction performance. This case study explores the use of an LSTM model to analyze historical Google stock data and predict future price trends.

Data Selection:

Google which is a subsidiary company of Alphabet Inc. is one of the most influential and famous technology companies in the world. Google was first largely recognized for its Internet search engine, but it has since expanded into a wide range of other fields and has grown to be a dominating player in the technology sector. The fact that Google is the most extensively used search engine in the world, that Google Cloud offers cloud infrastructure and services to companies, and that it is at the forefront of artificial intelligence and machine learning research all contribute to Google's enormous impact in the IT sector. As a result, Google's stock price is representative of the technology industry.

Data Preprocessing:

The dataset used in this study consists of Google's historical closing stock prices. Missing (null) values were removed to ensure data quality, and the data was normalized so that all features had equal importance during model training. The dataset was then divided into training (80%) and testing (20%) sets. Finally, the data was organized into sequences of 60 time steps, enabling the LSTM model to learn historical patterns and accurately predict future stock prices.

Results and Discussion:

The LSTM model produced predictions with low error, showing a close match between the predicted and actual Google stock prices. The results indicate that the model effectively captured overall market trends, making it useful for supporting investment decisions. Although minor fluctuations were not predicted accurately, the model demonstrated strong performance in forecasting stock price movements. Further improvements in data quality, feature selection, and model tuning could enhance prediction accuracy and reliability.

Conclusion:

This case study demonstrates that the LSTM machine learning model can effectively predict stock price trends using historical Google stock data. The model achieved high accuracy in capturing overall market movements, making it a valuable tool for supporting investment decisions. However, stock markets are influenced by unpredictable events, so perfect predictions are not possible. Future improvements in machine learning models, along with better data and ethical practices, can further enhance the accuracy and reliability of stock market forecasting.

References:

https://www.researchgate.net/publication/376887984_Machine_Learning_in_Finance_A_Case_Study_on_Forecasting_Google's_Stock_Price

1.3 Python Programming for Data Science

1. Basic Python Programs:

Python program to find FACTORIAL OF A NUMBER

```
|: def fact(n):  
    if(n==0 or n==1):  
        return 1  
    else:  
        return n*fact(n-1)  
n=int(input("Enter a number:"))  
result=fact(n)  
print(f"Factorial of {n} is {result}")
```

Enter a number: 5
Factorial of 5 is 120

```
|: import math  
n=int(input("Enter a number:"))  
result=math.factorial(n)  
print(f"Factorial of {n} is {result}")
```

Enter a number: 4
Factorial of 4 is 24

Python program to check if number is prime or not

```
n=int(input("Enter a number:"))  
if n>1:  
    for i in range(2,n):  
        if(n%i==0):  
            p=1  
            break  
        else:  
            p=0  
    if(p==0):  
        print(f"{n} is a prime number")  
    else:  
        print(f"{n} is not a prime number")
```

Enter a number: 8
8 is not a prime number

2. Control Structures:

Python program using if-else statement to determine whether the number is positive negative or zero

```
n=int(input("Enter a number:"))
if(n>0):
    print(f"{n} is a positive number")
elif(n==0):
    print(f"{n} is zero")
else:
    print(f"{n} is a negative number")
```

Enter a number: -9
-9 is a negative number

Python program to print first ten Fibonacci numbers

```
a=0
b=1
print("Fibonacci Series:")
for i in range(10):
    print(a, " ")
    c=a+b
    a=b
    b=c
```

Fibonacci Series:
0
1
1
2
3
5
8
13
21
34

3.Functions in Python:

Python program to calculate area of circle

```
r=int(input("Enter radius of circle"))
area=3.14*r*r
print(f"Area of circle is {area}")
```

```
Enter radius of circle 2
Area of circle is 12.56
```

Python program that takes a list of number and returns the largest number in the list

```
def large(lst):
    max=lst[0]
    for i in lst:
        if i>max:
            max=i
    return max
n=int(input("Enter number of elements of list:"))
lst=[]
for i in range(n):
    num=int(input("Enter the element:"))
    lst.append(num)
print("Largest element is:",large(lst))
```

```
Enter number of elements of list: 5
Enter the element: 2
Enter the element: 3
Enter the element: 89
Enter the element: 1
Enter the element: 43
Largest element is: 89
```

4.File Handling:

Python program to read a text file and count the numbers of words in it.

```
[2]: path="filefor1.3.txt"
a=open(path,"r")
text=a.read()
words=text.split()
count=len(words)
print("The numbers of word in the file=",count)
a.close()
```

```
The numbers of word in the file= 15
```

1.4 Python Data Structures

1. List and Tuples:

Write a Python program to find the second largest number in a list.

```
n=int(input("Enter the value of n:"))
print("Enter the elements:")
list1=[]
for i in range(0,n):
    a=int(input())
    list1.append(a)
print(list1)
largest=list1[0];
slargest=-1;
for i in range(0,n):
    if(list1[i]>largest):
        slargest=largest
        largest=list1[i]
    elif(list1[i]>slargest and list1[i]!=largest):
        slargest=list1[i]
print("The second largest element is ",slargest)
```

```
Enter the value of n: 6
Enter the elements:
1
3
2
4
6
7
[1, 3, 2, 4, 6, 7]
The second largest element is 6
```

Write a Python program to merge two tuples into a dictionary.

```
[8]: tuple1=("Cherry","Banana","Orange")
tuple2=("red","yellow","orange")
merge1=dict(zip(tuple1,tuple2))
print(merge1)

{'Cherry': 'red', 'Banana': 'yellow', 'Orange': 'orange'}
```

2. Dictionaries and Sets:

Practice Dictionary

```
[25]: thisdict={"Amazon":"brown","Blinkit":"yellow","Zomato":"red"}
print(thisdict["Blinkit"])
thisdict["Uber"]="black"
thisdict["Amazon"]="yellow"
thisdict.pop("Blinkit")
print(thisdict.items())
for x in thisdict:
    print(x)
for x,y in thisdict.items():
    print(x,y)
```

```
yellow
dict_items([('Amazon', 'yellow'), ('Zomato', 'red'), ('Uber', 'black')])
Amazon
Zomato
Uber
Amazon yellow
Zomato red
Uber black
```

Write a Python program to count the frequency of each word in a given string using a dictionary.

```
[28]: sentence="Good Morning Good Afternoon Good Evening Good Night"
words=sentence.split()
freq={}
for word in words:
    if word in freq:
        freq[word]+=1
    else:
        freq[word]=1
print(freq)

{'Good': 4, 'Morning': 1, 'Afternoon': 1, 'Evening': 1, 'Night': 1}
```

Practice sets

```
[30]: set1={1,1,1,1,2,3,4}  
print(set1)
```

```
{1, 2, 3, 4}
```

Write a Python program to find the union and the intersection of two sets.

```
[31]: set1={1,2,3,4,5}  
set2={3,4,5,6,7}  
common=set1.intersection(set2)  
print("Intersection:",common)  
uni=set1.union(set2)  
print("Union:",uni)
```

```
Intersection: {3, 4, 5}
```

```
Union: {1, 2, 3, 4, 5, 6, 7}
```

1.5 Introduction to Data Manipulation Libraries

1. Pandas Basics:

Pandas is a popular Python library used for data manipulation, analysis, and cleaning. It provides fast, flexible data structures like the Series and Data Frame to work with tabular data.

Core Data Structures

- Series: A 1D labelled array that can hold any data type (like a single column).
- Data Frame: A 2D labelled data structure with rows and columns (like an Excel sheet or SQL table).

Common Basic Operations

- Installation: Run `pip install pandas` in your terminal.

Create a DataFrame from a dictionary of lists.

```
[23]: import pandas as pd
student={"name":["Diya","Prachi","Khushi"],"age":[19,20,18]}
df=pd.DataFrame(student)
print(df,"\n")
print(df["name"],"\n")
print("Data from row no.3:\n",df.loc[2])
```

```
      name  age
0    Diya   19
1  Prachi   20
2  Khushi   18
```

```
0    Diya
1  Prachi
2  Khushi
Name: name, dtype: object
```

```
Data from row no.3:
      name  age
2  Khushi   18
Name: 2, dtype: object
```

Perform basic operations like selecting rows/columns, filtering, and sorting.

```
[27]: df_csv=pd.read_csv("file.csv")
print(df_csv.head())
print(df_csv.tail())
```

	Artist	Sex	Country	Language	Primary Genre	Artist Type \
0	Drake	Male	Canada	English	Hip-Hop	Solo
1	Taylor Swift	Female	United States	English	Pop	Solo
2	Bad Bunny	Male	United States	Spanish	Reggaeton	Solo
3	The Weeknd	Male	Canada	English	R&B	Solo
4	Justin Bieber	Male	Canada	English	Pop	Solo

	Total Streams (in millions)	Lead Streams (in millions) \
0	137492.1	94851.1
1	127861.0	126177.3
2	125899.8	82189.3
3	95703.2	77189.4
4	78104.4	48316.2

	Feature Streams (in millions)	Solo Streams (in millions)
0	42640.9	53323.9
1	1683.8	113421.5
2	43710.5	48887.0
3	18513.8	51020.0
4	29788.2	29638.3

	Artist	Sex	Country	Language	Primary Genre	Artist Type \
495	The Cure	Male	United Kingdom	English	Rock	Group
496	The Marías	Mixed	United States	English	Pop	Group
497	Limp Bizkit	Male	United States	English	Nu Metal	Group
498	Korn	Male	United States	English	Metal	Group
499	Omar Courtz	Male	United States	Spanish	Reggaeton	Solo

	Feature Streams (in millions)	Solo Streams (in millions)
495	3.3	6940.1
496	1261.4	5705.9
497	5.4	6806.0
498	2798.5	6414.9
499	20.3	1876.2

```
[29]: print(df_csv["Country"])
```

```
0          Canada
1    United States
2    United States
3          Canada
4          Canada
...
495  United Kingdom
496    United States
497    United States
498    United States
499    United States
Name: Country, Length: 500, dtype: object
```

```
[26]: print(df_csv.loc[3])
```

```
Artist          The Weeknd
Sex              Male
Country         Canada
Langugage       English
Primary Genre   R&B
Artist Type     Solo
Total Streams (in millions)  95703.2
Lead Streams (in millions)  77189.4
Feature Streams (in millions)  18513.8
Solo Streams (in millions)  51020.0
Name: 3, dtype: object
```

```
[28]: print(df_csv.iloc[3])
```



```
Artist          The Weeknd
Sex              Male
Country         Canada
Langugage       English
Primary Genre   R&B
Artist Type     Solo
Total Streams (in millions)  95703.2
Lead Streams (in millions)  77189.4
Feature Streams (in millions)  18513.8
Solo Streams (in millions)  51020.0
Name: 3, dtype: object
```

```
[8]: print(df_csv[df_csv["Sex"]=="Female"])
```

	Artist	Sex	Country	Language	Primary Genre	\
1	Taylor Swift	Female	United States	English	Pop	
5	Ariana Grande	Female	United States	English	Pop	
10	Rihanna	Female	Barbados	English	Pop	
11	Billie Eilish	Female	United States	English	Pop	
22	Nicki Minaj	Female	Trinidad and Tobago	English	Hip-Hop	
..	...	...	...	...	...	
470	Maiara & Maraisa	Female	Brazil	Portuguese	Sertanjeo	
476	MØ	Female	Denmark	English	Pop	
479	Laufey	Female	Iceland	English	Pop	
481	Nicki Nicole	Female	Argentina	Spanish	Latin	
483	Magenta	Female	Italy	Italian	Pop	

	Artist Type	Total Streams (in millions)	Lead Streams (in millions)	\
1	Solo	127861.0	126177.3	
5	Solo	68298.9	57097.5	
10	Solo	60670.7	44824.4	
11	Solo	58049.5	57160.1	
22	Solo	43330.3	16316.4	
..	...	...	...	
470	Group	7388.4	3471.7	
476	Solo	7335.2	6735.6	
479	Solo	7286.8	4703.7	
481	Solo	7256.2	2470.4	
483	Solo	7239.7	NaN	

```
[10]: print(df_csv.sort_values(by="Total Streams (in millions)"))
```

	Artist	Sex	Country	Language	Primary Genre	\
499	Omar Courtz	Male	United States	Spanish	Reggaeton	
498	Korn	Male	United States	English	Metal	
497	Limp Bizkit	Male	United States	English	Nu Metal	
496	The Marías	Mixed	United States	English	Pop	
495	The Cure	Male	United Kingdom	English	Rock	
..	...	...	...	...	...	
4	Justin Bieber	Male	Canada	English	Pop	
3	The Weeknd	Male	Canada	English	R&B	
2	Bad Bunny	Male	United States	Spanish	Reggaeton	
1	Taylor Swift	Female	United States	English	Pop	
0	Drake	Male	Canada	English	Hip-Hop	

	Artist Type	Total Streams (in millions)	Lead Streams (in millions)	\
499	Solo	6972.0	6956.0	
498	Group	6976.4	4188.4	
497	Group	7018.9	7018.4	
496	Group	7029.1	5774.4	
495	Group	7034.1	7035.0	
..	...	...	...	
4	Solo	78104.4	48316.2	
3	Solo	95703.2	77189.4	
2	Solo	125899.8	82189.3	
1	Solo	127861.0	126177.3	
0	Solo	137492.1	94851.1	

```
[12]: print(df_csv.sort_values(by="Total Streams (in millions)",ascending=False))
```

	Artist	Sex	Country	Language	Primary Genre	\
0	Drake	Male	Canada	English	Hip-Hop	
1	Taylor Swift	Female	United States	English	Pop	
2	Bad Bunny	Male	United States	Spanish	Reggaeton	
3	The Weeknd	Male	Canada	English	R&B	
4	Justin Bieber	Male	Canada	English	Pop	
..	...	...	...	...	...	
495	The Cure	Male	United Kingdom	English	Rock	
496	The Marías	Mixed	United States	English	Pop	
497	Limp Bizkit	Male	United States	English	Nu Metal	
498	Korn	Male	United States	English	Metal	
499	Omar Courtz	Male	United States	Spanish	Reggaeton	

	Artist Type	Total Streams (in millions)	Lead Streams (in millions)	\
0	Solo	137492.1	94851.1	
1	Solo	127861.0	126177.3	
2	Solo	125899.8	82189.3	
3	Solo	95703.2	77189.4	
4	Solo	78104.4	48316.2	
..	...	...	...	
495	Group	7034.1	7035.0	
496	Group	7029.1	5774.4	
497	Group	7018.9	7018.4	
498	Group	6976.4	4188.4	
499	Solo	6972.0	6956.0	

```
[15]: print(df_csv.loc[0:7])
```

	Artist	Sex	Country	Language	Primary Genre	Artist Type	\
0	Drake	Male	Canada	English	Hip-Hop	Solo	
1	Taylor Swift	Female	United States	English	Pop	Solo	
2	Bad Bunny	Male	United States	Spanish	Reggaeton	Solo	
3	The Weeknd	Male	Canada	English	R&B	Solo	
4	Justin Bieber	Male	Canada	English	Pop	Solo	
5	Ariana Grande	Female	United States	English	Pop	Solo	
6	Travis Scott	Male	United States	English	Hip-Hop	Solo	
7	Kanye West	Male	United States	English	Hip-Hop	Solo	

	Total Streams (in millions)	Lead Streams (in millions)	\
0	137492.1	94851.1	
1	127861.0	126177.3	
2	125899.8	82189.3	
3	95703.2	77189.4	
4	78104.4	48316.2	
5	68298.9	57097.5	
6	65349.7	35338.7	
7	65142.9	46558.4	

	Feature Streams (in millions)	Solo Streams (in millions)	\
0	42640.9	53323.9	
1	1683.8	113421.5	
2	43710.5	48887.0	
3	18513.8	51020.0	
4	29788.2	29638.3	
5	11201.4	41615.5	
6	30027.0	25309.9	
7	18584.5	28473.0	

2. NumPy Basics:

NumPy (Numerical Python) is an open-source Python library designed for numerical and scientific computing. It provides support for fast and efficient multidimensional arrays and matrices, making data processing easier than using regular Python lists. NumPy is widely used in data analysis, machine learning, artificial intelligence, and scientific research. It also serves as the foundation for many other Python libraries, such as Pandas, SciPy, and Matplotlib.

Create a NumPy array and perform element-wise operations.

```
[23]: import numpy as np
a=np.array([10,20,30,40,50])
a0=np.zeros((2,3))
a1=np.ones((2,2))
i=np.eye(3)
r=np.random.rand(2,2)
ar=np.arange(0,10,2)
l=np.linspace(1,10,4)
print(a)
print(a0)
print(a1)
print(i)
print(r)
print(ar)
print(l)
print(a.shape)
print(a.ndim)
print(a.size)
print(a.dtype)
```

```
[10 20 30 40 50]
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1.]
 [1. 1.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[[0.27057322 0.08550209]
 [0.31267652 0.34295617]]
[0 2 4 6 8]
[ 1.  4.  7. 10.]
(5,)
1
5
int64
```

```
[30]: A=np.array([4,9,2])
      B=np.array([1,3,5])
      print("Addition=",A+B)
      print("Substraction=",A-B)
      print("Multiplication=",A*B)
      print("Division=",A/B)
      print("Square=",A**2)
      print("Transpose of A=",np.transpose(A))
      print("Sum=",np.sum(A))
      print("Mean=",np.mean(A))
      print("Maximum=",np.max(A))
      print("Minimum=",np.min(A))
      print("Sort=",np.sort(A))
      p=np.array([[3,2],[1,4]])
      print(np.linalg.det(p))
      print(np.linalg.inv(p))
      print(np.linalg.eig(p))

Addition= [ 5 12  7]
Substraction= [ 3  6 -3]
Multiplication= [ 4 27 10]
Division= [4.  3.  0.4]
Square= [16 81  4]
Transpose of A= [4 9 2]
Sum= 15
Mean= 5.0
Maximum= 9
Minimum= 2
Sort= [2 4 9]
10.000000000000002
[[ 0.4 -0.2]
 [-0.1  0.3]]
EigResult(eigenvalues=array([2., 5.]), eigenvectors=array([[ -0.89442719, -0.70710678],
 [ 0.4472136 , -0.70710678]]))
```

Use NumPy to perform matrix multiplication.

```
[32]: a=np.array([[1,2],[3,4]])
      b=np.array([[5,6],[7,8]])
      c=a@b
      print(c)
      d=np.dot(a,b)
      print(d)
      e=np.matmul(a,b)
      print(e)

[[19 22]
 [43 50]]
[[19 22]
 [43 50]]
[[19 22]
 [43 50]]
```

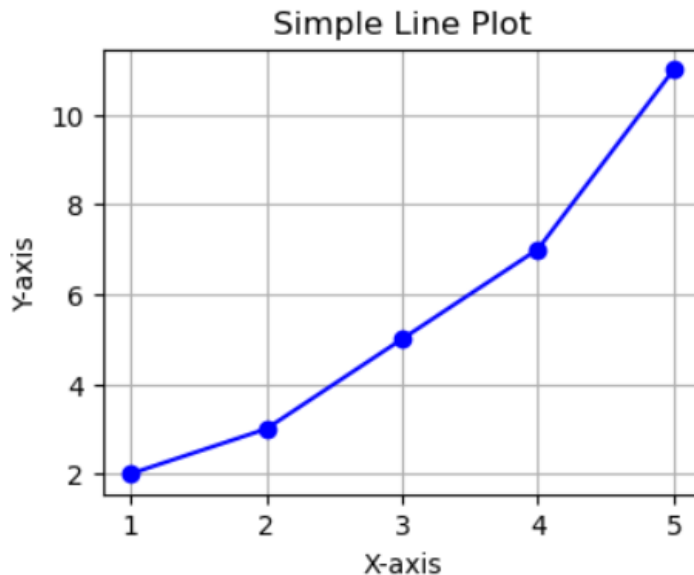
3. Data Visualization with Matplotlib:

Matplotlib is a Python library used for creating static, animated and interactive data visualizations. It is built on the top of NumPy and it can easily handles large datasets for creating various types of plots such as line charts, bar charts, scatter plots, etc.

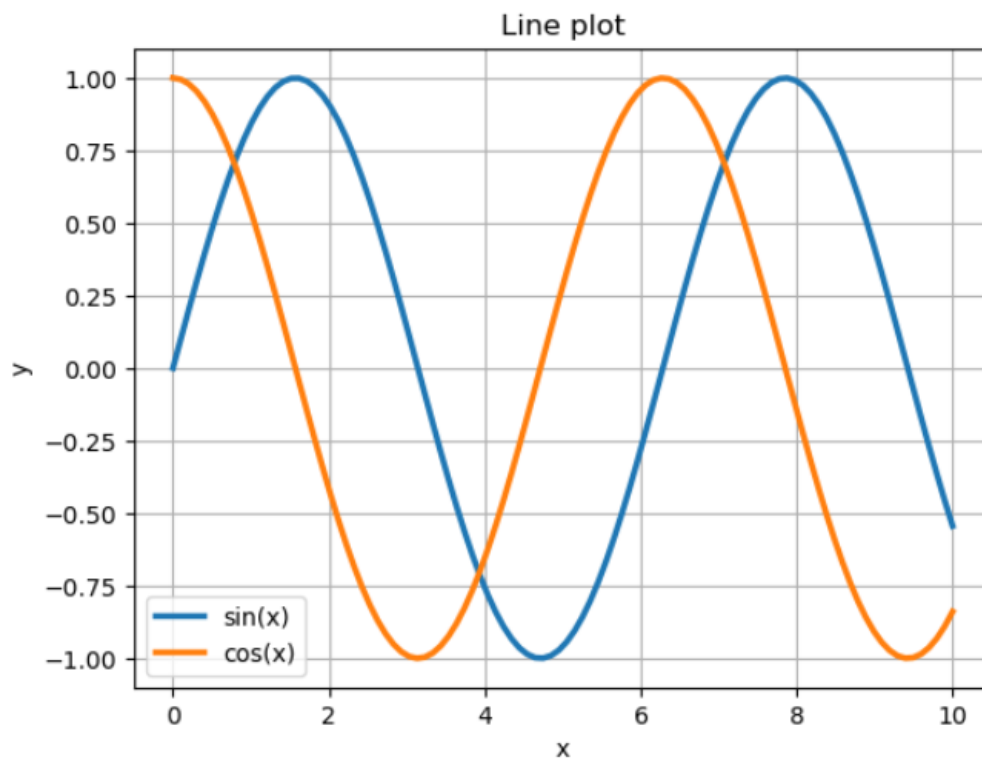
Create a simple line plot using Matplotlib.

```
[5]: import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[2,3,5,7,11]

plt.figure(figsize=(4,3))
plt.plot(x,y,marker="o",linestyle="-",color="blue")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Simple Line Plot")
plt.grid(True)
plt.show()
```



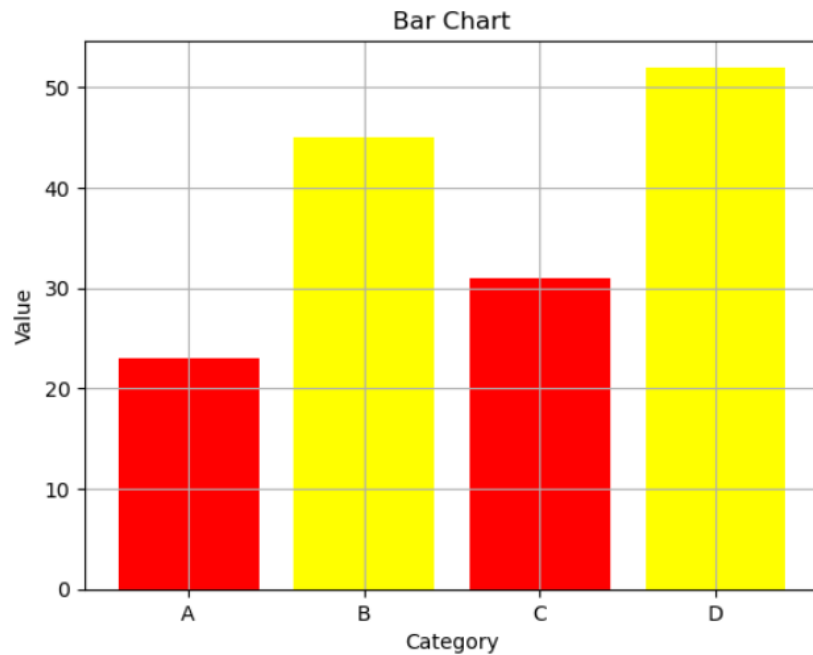
```
[7]: import matplotlib.pyplot as plt
import numpy as np
x=np.linspace(0,10,100)
plt.plot(x,np.sin(x),label="sin(x)",linewidth=2.5)
plt.plot(x,np.cos(x),label="cos(x)",linewidth=2.5)
plt.title("Line plot")
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.grid()
plt.show()
```



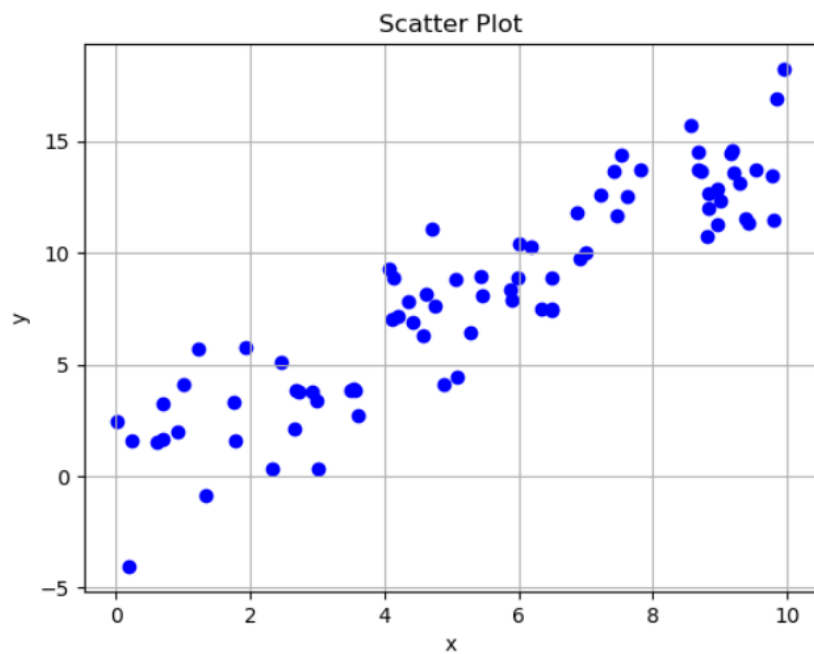
Create a bar chart to visualize categorical data.

```
[10]: import matplotlib.pyplot as plt

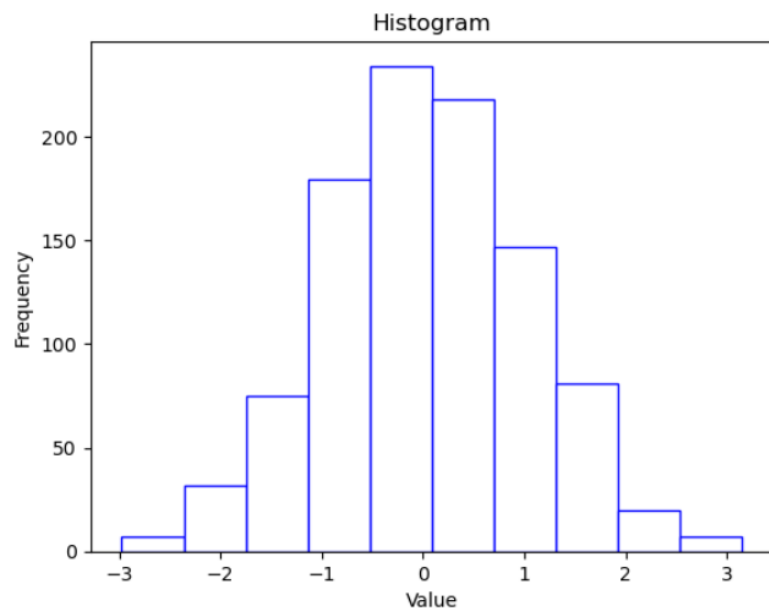
categories = ["A", "B", "C", "D"]
values = [23, 45, 31, 52]
plt.bar(categories, values, color=["red", "yellow", "red", "yellow"])
plt.title("Bar Chart")
plt.xlabel("Category")
plt.ylabel("Value")
plt.grid()
plt.show()
```



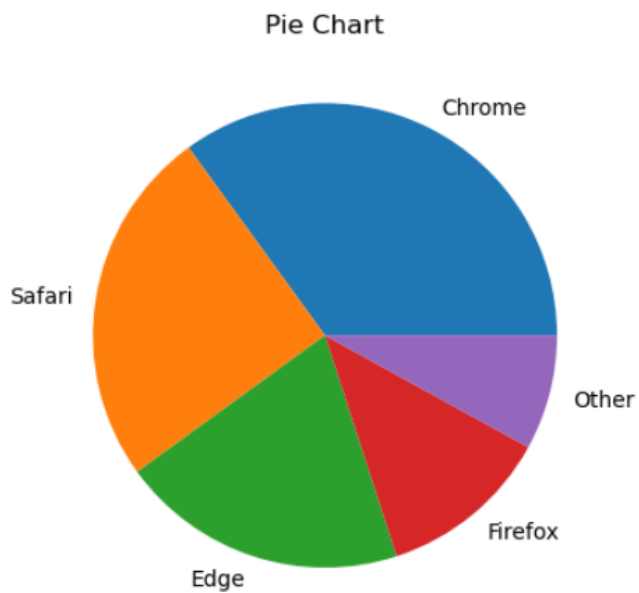
```
[12]: x=np.random.rand(80) * 10
      y=x*1.5 + np.random.randn(80) * 2
      plt.scatter(x,y,color="b",marker="o")
      plt.title("Scatter Plot")
      plt.xlabel("x")
      plt.ylabel("y")
      plt.grid()
      plt.show()
```



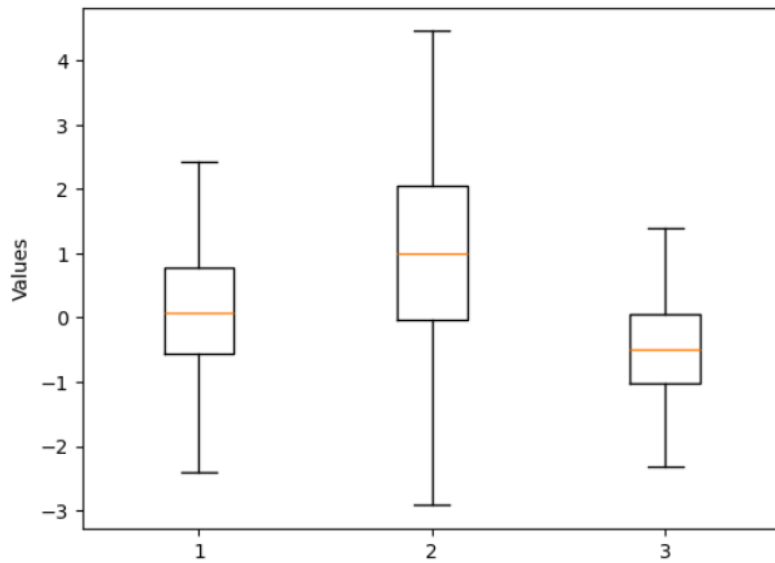
```
[21]: data = np.random.normal(0,1,1000)
plt.hist(data,color="w",edgecolor="b")
plt.title("Histogram")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```



```
[24]: sizes=[35,25,20,12,8]
labels=["Chrome","Safari","Edge","Firefox","Other"]
plt.pie(sizes,labels=labels)
plt.title("Pie Chart")
plt.show()
```



```
[31]: data=[np.random.normal(0,1,200),
          np.random.normal(1,1.5,200),
          np.random.normal(-0.5,0.8,200)]
plt.boxplot(data)
plt.ylabel("Values")
plt.show()
```



```
[47]: import pandas as pd
data=pd.read_csv("train.csv")
x=data['PassengerId']
y=data['Age']
plt.plot(x,y,marker='o',linestyle='-')
plt.title("Titanic")
plt.xlabel("ID")
plt.ylabel("Age")
plt.show()
```

